

No Gender Differences in Long-Term Survival after Brachytherapy of 1,541 Patients with Uveal Melanoma

Gustav Stålhammar^{a–c} Thonnie Rose See^c Maria Fili^a Stefan Seregard^{a, b}

^aSt. Erik Eye Hospital, Stockholm, Sweden; ^bDepartment of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden; ^cDepartments of Ophthalmology and Pathology, Emory University School of Medicine, Atlanta, GA, USA

Keywords

Gender · Brachytherapy · Uveal melanoma · Survival

Abstract

Background: In several malignancies, gender-based survival differences after specific therapeutic interventions have been demonstrated. It is not known whether such differences exist after plaque brachytherapy of uveal melanoma. **Methods:** All patients who received brachytherapy for uveal melanoma at St. Erik Eye Hospital from November 1, 1979 through November 20, 2017 were included ($n = 1,541$). Retrospective data were retrieved including baseline patient and tumor characteristics, brachytherapy nuclide (ruthenium-106 or iodine-125), radiation dose, treatment duration, tumor relapses, date of metastasis, and cause of death. **Results:** A total of 775 men and 766 women were treated with plaque brachytherapy. There were no significant differences between the genders in baseline characteristics, treatment, or follow-up. Men and women had similar rates of tumor relapses, hazard for repeated brachytherapy (men vs. women 0.8, $p = 0.47$), enucleation-free survival, and survival after de-

tection of metastasis. Five-, 10-, and 15-year melanoma-related mortality was 14, 24, and 27% for men and 15, 26, and 32% for women, respectively. There were no significant differences in hazard for melanoma-related mortality (men vs. women 0.9, $p = 0.32$), median Kaplan-Meier disease-specific survival (men 18.2 years, women 15.5 years, $p = 0.22$), or median overall survival (men 13.5 years, women 12.6 years, $p = 0.60$). **Conclusion:** There are no relevant differences between men and women in ocular or patient survival after brachytherapy for uveal melanoma.

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Introduction

Uveal melanoma is the most common primary intraocular malignancy in adults, with a 15-year disease-related mortality of 45% [1, 2]. Plaque brachytherapy is the standard evidence-based treatment for medium-sized primary tumors (2.5–10 mm in thickness, <15 mm in diameter) [3]. Smaller tumors are observed for growth or treated with transpupillary thermotherapy, while eyes

with larger tumors are generally enucleated [4, 5]. Men have less symptoms and larger and more posterior tumors at presentation [6, 7]. Likely a consequence of this, men had earlier metastases in some studies [6]. Nonetheless, similar incidence, age at presentation, recurrence rates, and survival between the genders have been reported in several [7–9] but not all [6, 10, 11] patient series.

In terms of outcomes after specific interventions, women have significantly better overall and progression-free survival than men after yttrium-90 brachytherapy of uveal melanoma liver metastases, adjusted for age, patient performance status, tumor function parameters, and presence of extrahepatic disease [12]. The mechanism behind this difference is elusive. Outside ophthalmology there are clues pointing to a gender-based difference in radiosensitivity: it has been shown that after exposure to a 100 mSv dose to the chest, women have more than a doubled lifetime risk of developing lung cancer compared to men [13]. After a dose of 2 Gy, the survival proportion of strains of fibroblasts derived from women is significantly lower than that of strains derived from men [14]. On the other hand, there are also indications of differences between the genders in the likelihood of undergoing treatment and delay to treatment in several malignancies. Women with bladder cancer are more likely to undergo radical cystectomy but have worse disease-specific survival than men [15]. Women with colon cancer are more likely to undergo surgery as opposed to chemotherapy [16], but have higher mortality if diagnosed after the age of 65 years [17]. And women with non-Hodgkin's lymphoma have longer prehospital, referral, and treatment delay [18].

Corresponding gender differences after plaque brachytherapy treatment of primary uveal melanoma cannot be ruled out based on similar survival when all treatment modalities are included, or differences in survival after brachytherapy of metastases. To the best of our knowledge, no previous study has specifically investigated such differences. We therefore see an opportunity to review our plaque brachytherapy experience since 1979, with special regard to patient and tumor characteristics at baseline and long-term survival.

Methods

All patients who received ruthenium-106 or iodine-125 brachytherapy for choroidal melanoma at the Pathology and Oncology Service, St. Erik Eye Hospital from November 1, 1979 through November 20, 2017 were considered for the study ($n = 1,857$). In this period, our institution was the only one in the country performing

plaque brachytherapy. Retrospective data were retrieved from digitalized clinical records and from our digital Brachytherapy of Uveal Melanoma directory, including information on gender, age at diagnosis, pretreatment visual acuity, symptom duration before presentation, type of symptoms, tumor origin (choroid, ciliary body, or iris), tumor thickness and diameter, AJCC T-category, brachytherapy nuclide (ruthenium-106 or iodine-125), prescribed radiation dose, treatment duration, transpupillary thermotherapy treatment before or after brachytherapy, enucleation after brachytherapy, date of metastasis, and follow-up years. The date of death was retrieved from the National Mortality Registry. This registry is administered centrally and supplied with data from all health-care institutions in the country, including classifications of cause of death. As it has been shown that misclassifications of the actual cause of death are common, primarily between cutaneous and choroidal melanoma, we crosschecked the National Mortality Registry's data with our own clinical records of patients' metastases [19]. If a patient had developed one or several metastases of choroidal melanoma and later died from metastasized melanoma without specified primary location, we considered the actual cause of death to be metastasized choroidal melanoma.

In review of the data in the Brachytherapy of Uveal Melanoma directory, 295 patients were excluded as they had actually undergone primary enucleation rather than plaque brachytherapy (indications were tumor size too large and/or inaccessible for satisfactory plaque placement [$n = 274$], optic disc invasion [$n = 13$], patient preference [$n = 3$], extensive inflammation [$n = 1$], vitreous bleeding [$n = 1$], extensive retinal detachment [$n = 1$], extrascleral tumor growth [$n = 1$], and concurrent prostate cancer metastasis in the same eye [$n = 1$]). Fifteen patients were excluded as they did not have treatment dates specified, and 5 were excluded because the site of origin for their tumor was the iris as opposed to the ciliary body or choroid. One patient was excluded since no measurement of the tumor's thickness was available, yielding a total of 1,541 patients eligible for analysis.

Our protocol for ruthenium-106 brachytherapy, introduced in 1979, allows for a maximum scleral dose of 1,500 Gy. This makes it possible to prescribe 100 Gy at the tumor apex + 1 mm for tumors with a thickness of up to 7 mm. For iodine-125, introduced in 1999, the treatment was planned for 100 Gy until 2003, after which it was lowered to 80 Gy due to concern for surrounding tissues.

All ruthenium plaques were of the CCA, CCB, CCX, CCZ, COB, CIA, or CIB types (Eckert & Ziegler BEBIG GmbH, Berlin, Germany). Iodine plaques are custom-made by a goldsmith in 18-carat gold alloy to replicate the CCA, CCB, CCC, CIB, COB, and CCZ types, with iodine seeds (Eckert & Ziegler BEBIG GmbH) glued to the concave surface. Source specification data from the manufacturer were verified by medical physicists at the Karolinska University Hospital, Stockholm. All patients had the tumor thickness measured preoperatively by standardized A- and B-scan ultrasonography. Surgery was usually performed under general anesthesia and included transillumination followed by plaque positioning with a minimum 2-mm margin around the tumor. Some tumors close to the fovea had eccentric plaque fixation and adjunctive laser therapy because of the compromised radiotherapy margin. Juxtapapillary melanomas were either treated with a notched plaque or standard plaque combined with adjunctive laser therapy.

Regular follow-up was scheduled at 1, 3, 6, and 12 months after therapy, and then annually or semi-annually if local control had been achieved. At each visit, best-corrected visual acuity, intraocu-

lar pressure, slit-lamp examination, biomicroscopy, indirect and direct ophthalmoscopy, transillumination, ultrasonography with standardized A- and B-scan, and fundus photography were undertaken. When tumor regression is deemed insufficient or there is tumor progression 6 months after brachytherapy or later, a second round of brachytherapy can be considered. The reason for an individual patient's repeated brachytherapy or post-brachytherapy enucleation is usually described in the clinical record. At each follow-up, the visual acuity was measured on a decimal scale. Semi-annual screening for liver metastases by ultrasonography or computed tomography for 5 years after choroidal melanoma diagnosis was performed in accordance with the wishes of the patient and in collaboration with a physician at the referring hospital.

After a few years, patients living outside the Stockholm area may be referred back to their home clinic. This does not affect data on repeated brachytherapy as these are exclusively performed at our institution. Similarly, data from enucleations will not be affected as all specimens are sent for histopathological examination at our institution. The number of tumor relapses can consequently be accounted for.

Statistical Analysis

In power calculations, we considered recommendations for equivalence testing [20]. These state that for 90% power, the sample size for each arm should be at least 541 patients, given a worst-case scenario probability of absent outcome (no metastasis or death) of $p_C = 0.5$ and a smallest detectable difference between the groups of $\delta = 10\%$. Reversely, in our sample size of 775 men and 766 women, we would be able to detect differences of down to 8–9%, with 90% power. The Mann-Whitney U test was used for tests of the null hypothesis in nonparametrically distributed group level data. The Pearson χ^2 test was used for comparisons of proportions in two-by-two tables. For analyses of hazard for post-brachytherapy enucleations, repeated brachytherapy, and melanoma-related mortality, bi- and multivariate Cox regressions with tumor thickness and diameter as covariates were used. To test whether our survival data met the proportional hazard assumption, we built a Cox regression model with a time-dependent versus a time-independent treatment variable (man or woman). In reports of disease-specific and overall survival, Kaplan-Meier statistics were computed and the log rank (Mantel-Cox) test of equality of survival distributions was applied. Equivalence between men and women was tested as the relative risk for melanoma-related death, with modified two-sided 95% CIs as for binary endpoints [20]. To allow for statistical comparison of AJCC T-category [21], T-categories were codified as T1 = 1, T2 = 2, T3 = 3, and T4 = 4. Average visual acuities were calculated on -1 LogMar converted values, which were then converted back to the decimal scale. Data on actual brachytherapy duration (days, hours) measured at the time of plaque removal and planned brachytherapy duration required to achieve the prescribed tumor apex dose based on preoperative dosimetry were available. To allow for a comparison of deviations from the prescribed dose, we used the following formula: estimated actual dose = actual brachytherapy duration / planned brachytherapy duration $\times$ prescribed dose.

Enucleation-free survival was defined as the time in years from brachytherapy to enucleation or, in the absence of enucleation, to last follow-up. Disease-specific survival was defined as the proportion of patients who had not died from metastatic melanoma before the end of follow-up and overall survival as the proportion of

patients who were still alive, according to data from digitalized clinical records and the National Mortality Registry. Differences with a p value < 0.05 were considered significant, all p values being two-sided. All statistical analyses were performed using SPSS statistics version 25 (IBM, Armonk, NY, USA).

Results

Descriptive Statistics

From 1979 to 2017, 775 men and 766 women were treated with plaque brachytherapy. Of these, 761 tumors in men and 753 tumors in women originated from the choroid, and 14 and 12, respectively, from the ciliary body. Of these patients, 626 men and 619 women were treated with ruthenium-106 and 149 men and 147 women with iodine-125. There were no significant differences between the genders in age at diagnosis, pretreatment visual activity, symptom duration before presentation, types of symptoms at presentation, tumor origin, tumor thickness, tumor diameter, AJCC T-category, brachytherapy nuclide used (ruthenium-106 or iodine-125), prescribed radiation dose, intended treatment duration, actual treatment duration, proportion of transpupillary thermotherapies before or after brachytherapy, or follow-up years (Table 1). Men had borderline statistically significantly thicker and wider tumors than women at presentation ($\chi^2 p = 0.06$ and $p = 0.08$, respectively). Our data on time from brachytherapy to death or last follow-up met the proportional hazard assumption ($p = 0.83$).

Differences in Planned versus Actual Treatment Duration

The mean planned treatment duration was 93.4 h for men and 96.6 h for women. The deviation from this, measured at the actual time of removal of the plaque, was similar between the genders (Table 1). In a bivariate Cox regression with tumor thickness as a covariate, there were no significant differences in hazard ratio (HR) for melanoma-related mortality in patients who received > 90 Gy (reference category), 80–90 Gy (HR 1.1, 95% CI 0.7–1.7, $p = 0.072$), 70–80 Gy (HR 0.9, 95% CI 0.6–1.5, $p = 0.081$), or < 70 Gy (HR 1.2, 95% CI 0.9–1.7, $p = 0.22$).

Rates of Repeated Brachytherapy

Data on rates of repeated brachytherapy were available for the years 2000–2017. During this period, 21 men and 27 women underwent a second round of brachytherapy (Pearson $\chi^2 p = 0.36$). None of them had their first brachytherapy before the year 2000. Among men, the

Table 1. Demographics and clinical features of patients and tumors included in this study

	Males (<i>n</i> = 775)	Females (<i>n</i> = 766)	<i>p</i> value ^a
Mean age at diagnosis, years	63 (14)	63 (14)	0.34
Mean pretreatment visual acuity	0.3 (0.3)	0.3 (0.2)	0.80
Mean symptom duration before presentation, months	0.7 (2.6)	0.7 (2.1)	0.81
Symptoms at presentation			
Reduced visual acuity	219 (28%)	224 (29%)	0.88
Shadow in visual field	92 (12%)	106 (14%)	0.33
Pain	4 (0.5%)	4 (0.5%)	0.97
Metamorphopsias	12 (2%)	8 (1%)	0.53
Photopsias and/or myodesopsia	72 (9%)	67 (9%)	0.56
Tumor origin			0.72
Choroid	761 (98%)	753 (98%)	
Ciliary body	14 (2%)	12 (2%)	
Mean tumor thickness, mm	5.1 (2.3)	5.2 (2.3)	0.06
Mean tumor diameter, mm	10.4 (3.5)	10.7 (3.5)	0.08
T-category			0.99
1	427 (55%)	404 (53%)	
2	235 (30%)	247 (32%)	
3	104 (13%)	104 (14%)	
4	9 (1%)	11 (1%)	
Brachytherapy nuclide			0.99
Ruthenium-106	626 (81%)	619 (81%)	
Iodine-125	149 (19%)	147 (19%)	
Mean prescribed radiation dose, Gy			
Tumor apex + 1 mm	94.5 (12.9)	94.5 (13.2)	0.71
Mean treatment duration, h			
Intended	93.4 (76.8)	96.6 (66.9)	0.65
Actual	92.5 (54.6)	95.6 (54.5)	0.24
TTT ^b	95 (12%)	90 (12%)	0.75
Mean follow-up, years ^c	6.3 (3.0)	6.2 (3.1)	0.89

Values are presented as mean (standard deviation) or *n* (%). ^a Pearson χ^2 *p* for two-by-two categorical tables, Mann-Whitney *U* *p* for continuous variables, and Cochran-Armitage trend test for >2 ordered factors. ^b Proportion of patients treated with transpupillary thermotherapy at any point prior to or after brachytherapy. ^c Mean follow-up was defined as the mean time in years from brachytherapy to death or to the last time patients were seen alive.

main reason for the repeated brachytherapy was tumor relapse in 19 and insufficient regression in 2. Among women, the reason for the repeated brachytherapy was tumor relapse in 23, increasing retinal detachment suspected to be caused by a tumor relapse in 1, and insufficient regression in 3. The mean interval from first to second brachytherapy was 2.2 years for men (SD 1.9) and 2.8 years for women (SD 2.2) (Mann-Whitney *U* *p* = 0.45). In a multivariate Cox regression with tumor thickness and diameter as covariates, there were no significant differences in HR for repeated brachytherapy between men and women (HR men vs. women 0.8, 95% CI 0.5–

1.4, *p* = 0.47). Of the 48 men and women who underwent a repeated brachytherapy, 17 (35%) eventually died of uveal melanoma.

Enucleation-Free Survival

One hundred and seven (14%) men and 111 (14%) women underwent enucleation at any point after brachytherapy. Among men, the main reason for the post-brachytherapy enucleation was tumor relapse in 97, total retinal detachment in 4, radiation retinopathy in 2, chronic vitreous bleeding in 1, tumor necrosis in 1, patient preference in 1, and chronic choroidal detachment

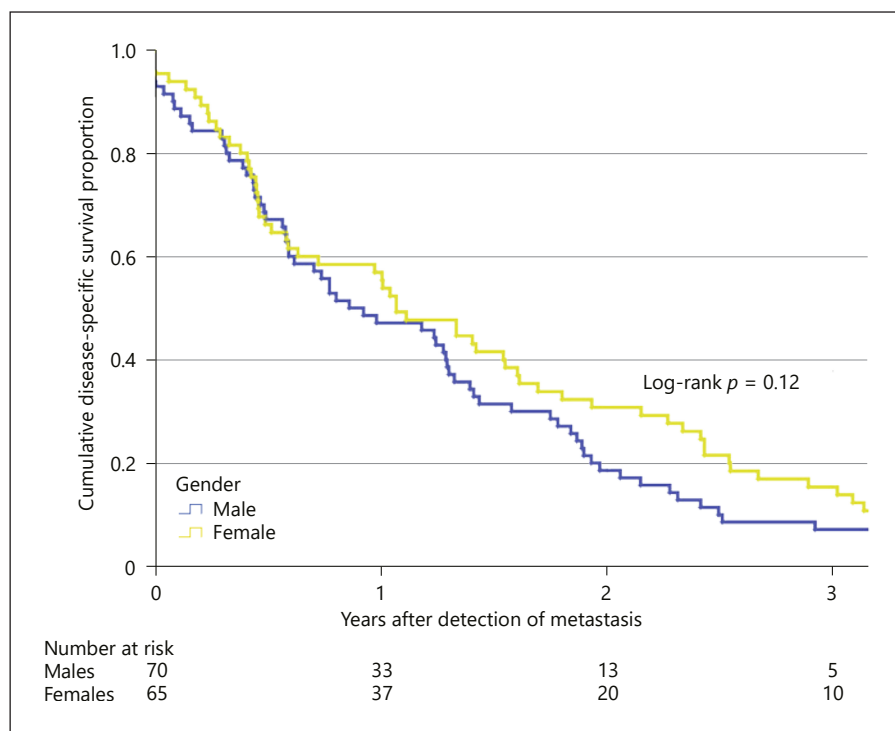


Fig. 1. Kaplan-Meier cumulative disease-specific survival proportion after detection of metastasis. Median survival was 0.9 years for men (95% CI 0.3–1.4) and 1.1 years for women (95% CI 0.7–1.4, log-rank $p = 0.12$).

in 1. Of the 107 men who underwent post-brachytherapy enucleation, 2 had first undergone a repeated brachytherapy. Among women, the main reason for the repeated brachytherapy was tumor relapse in 98, total retinal detachment in 3, chronic vitreous bleeding in 3, insufficient regression in 2, radiation retinopathy in 1, tumor necrosis in 1, patient preference in 1, high intraocular pressure in 1, and chronic uveitis in 1. Of the 111 women who underwent post-brachytherapy enucleation, 8 had first undergone a repeated brachytherapy. Added together, 116 men (15%) and 121 women (16%) had tumor relapse requiring intervention (repeated brachytherapy and/or enucleation) after brachytherapy (Pearson $\chi^2 p = 0.65$).

Men and women had similar Kaplan-Meier enucleation-free survival after brachytherapy (log-rank $p = 0.86$). Of the 218 men and women who underwent a post-brachytherapy enucleation, 72 (33%) eventually died of uveal melanoma.

Survival after Metastasis

Seventy men and 65 women had the date of detection of metastasis and survival thereafter specified. After detection of metastasis, the Kaplan-Meier median survival was 0.9 years for men (95% CI 0.3–1.4) and 1.1 years for women (95% CI 0.7–1.4) (log-rank $p = 0.12$, Fig. 1).

Melanoma-Related Mortality

Five-, 10-, and 15-year melanoma-related mortality was 14, 24, and 27% for men and 15, 26, and 32% for women, respectively.

In a bivariate Cox regression with tumor thickness and gender as covariates, there were no significant differences in HR for melanoma-related mortality between men and women (HR men vs. women 0.9, 95% CI 0.7–1.1, $p = 0.32$) (Fig. 2).

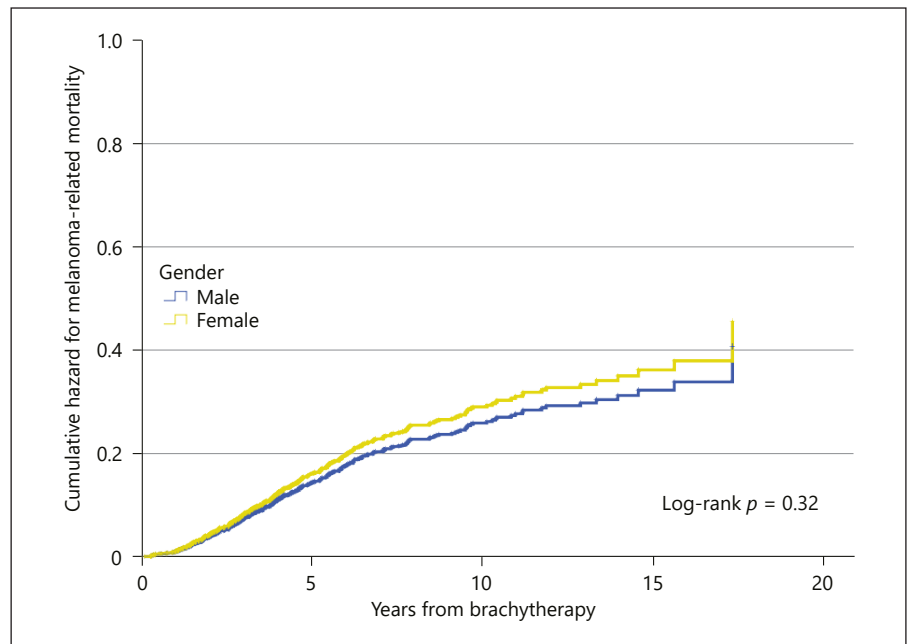
Similarly, there were no major differences in median Kaplan-Meier disease-specific survival (men 18.2 years, 95% CI 13.4–22.9; women 15.5 years, 95% CI 13.8–17.1; $p = 0.22$) (Fig. 3a) or overall survival (men 13.5 years, 95% CI 11.7–15.3; women 12.6 years, 95% CI 10.7–14.6; $p = 0.60$) (Fig. 3b).

In a test of equivalence, the relative risk of melanoma-related death for men versus women was 0.9, with a modified 95% CI of 0.7–1.0.

Discussion

In this study, there were no significant differences between the genders in tumor or patient characteristics at presentation or in outcome after brachytherapy in terms of deviations from planned treatment, enucleation-free survival, tumor relapses, metastasis-free survival, survival

Fig. 2. Bivariate Cox regression cumulative hazard for melanoma-related mortality of men versus women of 0.9 (95% CI 0.7–1.1, log-rank $p = 0.32$) after brachytherapy. Gender and tumor thickness were entered as covariates.



al after detection of metastasis, thickness-controlled hazard for melanoma-related mortality, or long-term Kaplan-Meier survival.

Similarly, the hazard for repeated brachytherapy was comparable between the genders. The event rate was however limited, inducing a wide CI and a result to be interpreted with caution.

Men had borderline larger tumors at presentation, comparable with a previous study by Damato and Coupland [7] in which men from the British Isles had significantly thicker and wider tumors. Andreoli et al. [8] and Gamel et al. [9] found no survival difference between men and women in databases of 7,043 and 740 American patients, respectively, without specific regard to treatment modality. In contrast, a study by Park et al. [10] found better survival rates in women than in men in a cohort of South Korean patients controlled for age, year of diagnosis, tumor site, and diagnostic verification method, but not tumor size. Zloto et al. [6] found better melanoma-related survival rates in women than in men with similar baseline characteristics in a report from Israel in which patients had undergone ruthenium-106 brachytherapy (79% of the women and 75% of the men), enucleation, local resection, or proton beam irradiation. Although baseline patient and tumor characteristics were similar between the genders in this study, comparisons of tumor size were on a small, medium, or large classification level without regard to millimeter-by-mil-

limeter thickness and diameter differences, which have been shown to be prognostically highly relevant [22, 23]. Rietschel et al. [11] showed that female sex correlated independently with better survival in metastatic uveal melanoma in a multivariate analysis of American patients.

One can consequently argue that there is no previous evidence of gender-based survival differences in primary uveal melanoma when prognostic variables such as tumor size are fully corrected for.

Our 5- and 10-year melanoma-related mortality rates of 14 and 24% for men and 15 and 26% for women essentially matched the COMS study of medium-sized tumors: the 5-year cumulative mortality rate with confirmed or suspected melanoma metastasis was 13% in both the enucleation and brachytherapy arms, and the 10-year mortality was 21% in the enucleation and 22% in the brachytherapy arm. On the other hand, our 15-year melanoma-related mortality of 27% for men and 32% for women is significantly lower than the 45% reported by Kujala et al. [2] and significantly higher than the 17, 14, and 18% for patients <20 years, 21–60 years, and >60 years reported by Shields et al. [24].

In addition to wide CIs for some of our comparisons, e.g., repeated brachytherapy and disease-specific survival, the limitations of this study include the potential for unmeasured covariates in regression analyses and the risk that our directory may misclassify or not capture all

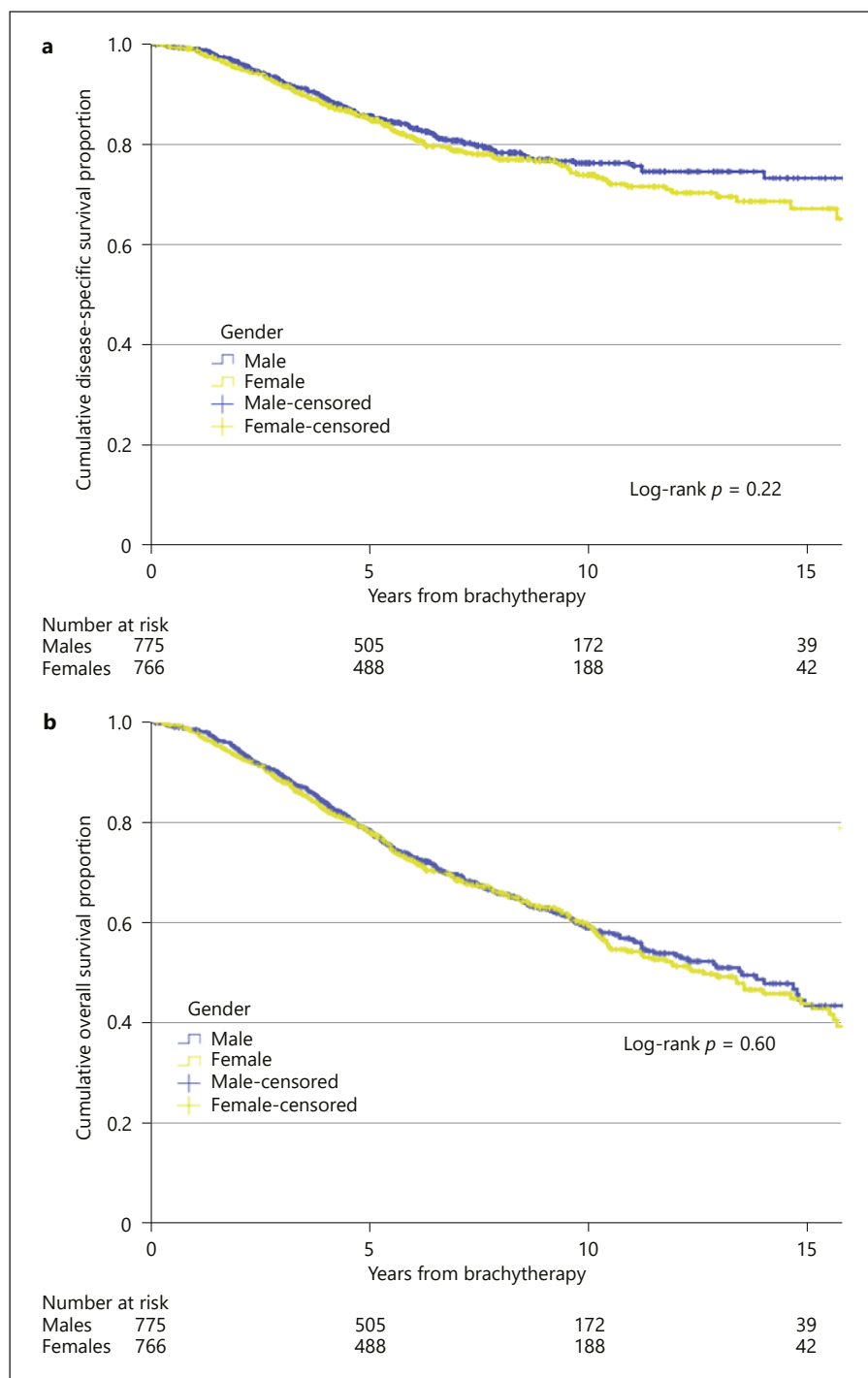


Fig. 3. a Kaplan-Meier cumulative disease-specific survival proportion after brachytherapy. Median survival in men 18.2 years (95% CI 13.4–22.9) and in women 15.5 years (95% CI 13.8–17.1, log-rank $p = 0.22$). **b** Kaplan-Meier cumulative overall survival proportion after brachytherapy. Median survival in men 13.5 years (95% CI 11.7–15.3) and in women 12.6 years (95% CI 10.7–14.6, log-rank $p = 0.60$).

events. Primarily, uveal melanoma-related deaths are at risk for being misclassified as skin melanoma-related deaths. We therefore sought to minimize such errors by cross-reference to our own clinical records. Further, the data are retrospective in nature and generated at one institution only, restricting generalizability.

Still, we do believe our results add to indications from related studies that there are no relevant differences in ocular or patient survival after brachytherapy of uveal melanoma between men and women. Gender-based differences in uveal melanoma radiosensitivity or treatment practices are thereby unlikely.

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Statement of Ethics

The study adhered to the tenets of the Declaration of Helsinki and the protocol was approved by the regional ethical review board of Stockholm.

Disclosure Statement

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